

Visual Augmentation of Audio Semantics for Accessibility

Working Knowledge Base: Literature Review, Design Notes & Project Plan

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Abstract

This document is the living record of the MSc project. It summarises the refined project definition, a grounded literature review across the eight areas named in the proposal, concrete model and dataset recommendations with hardware implications, the evaluation design, and a two-month work plan. It also logs the key decisions taken so far so that the full history of the project is recoverable from a single place. Sections are updated as the work progresses; a dated changelog is kept in Section 13.

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1 Executive Summary

The project builds a system that makes the *non-speech* information in a video’s soundtrack accessible to deaf and hard-of-hearing (DHH) viewers. Subtitles and sign-language interpretation convey *speech*, but they largely discard ambient and environmental sound — flowing water, approaching footsteps, a distant siren, rain, birds, a reacting crowd, machinery — which carries real meaning and atmosphere. The system detects this missing audio semantics and renders it as **complementary visual augmentations displayed alongside the original video**, supplementing (not replacing) the video and its captions.

The distinctive idea is **cross-modal gating**: because the *video* is also analysed, the system augments only what the audio conveys but the picture does *not* already show. A siren whose source is already on screen needs no augmentation; a siren from off-screen is exactly what is worth illustrating. This “reason about the gap between what is heard and what is seen” framing is the core contribution and the axis on which the method is compared to baselines that ignore the visual stream.

No models are trained. The contribution is (i) the *task formulation*, (ii) an inference pipeline that orchestrates existing open-source foundation models, (iii) a curated benchmark, and (iv) an automatic evaluation protocol. Work is scoped to roughly two months on a modest local GPU (GTX 1060 6 GB) with a university GPU requested for the diffusion and large-VLM stages.

Status at time of writing. Project definition refined and locked; literature review drafted (this document); model/dataset shortlist and hardware plan produced; two-month schedule produced. Implementation of the new pipeline not yet started — a prior code skeleton (FFmpeg audio extraction, Whisper ASR wiring, modular stage architecture) is reusable.

2 Project Definition and Scope

2.1 Problem statement

Given a short video clip (10–20s) with synchronised audio and video, identify semantic information in the audio — with emphasis on ambient and environmental sound — that is *insufficiently conveyed by the visual stream*, and generate complementary visual augmentations that communicate it to a DHH viewer.

2.2 Inputs and outputs

Input: a short clip with synchronised video + audio. Candidate sources: movie/TV scenes, documentaries, real-world recordings. The audio may contain speech, ambient sound, environmental sound, discrete acoustic events, and background music. **Output:** a set of complementary visual augmentations shown alongside the original video. Static, storyboard-style images first; short generated video segments are a stretch goal.

2.3 The seven-stage pipeline

The proposal defines a modular inference pipeline in which each stage can be evaluated or replaced independently:

1. **Audio extraction** — FFmpeg.
2. **Video understanding** — what is already visible (Qwen2.5-VL / LLaVA / InternVL).
3. **Speech recognition** — transcribe speech (Whisper Large V3 / Qwen2-Audio).
4. **Audio event detection** — non-speech sounds (BEATs / PANNs / AudioSet models).
5. **Cross-modal semantic analysis** — fuse scene context, speech and audio events; decide which cues matter, are *not* already visible, and how to depict them (Qwen3 / Llama 3.1).
6. **Visual augmentation generation** — FLUX.1 / Stable Diffusion XL.
7. **Evaluation.**

2.4 Scope boundaries and the “no training” constraint

Off-the-shelf open-source foundation models are glued together; nothing is trained or fine-tuned. The intellectual contribution is task formulation, cross-modal orchestration (especially stage 5), benchmark construction, and evaluation methodology.

2.5 Note on the change of direction

An earlier version of this project illustrated the *speech* of news clips for Deaf viewers. The current proposal is a deliberate pivot: the star is now *non-speech ambient/environmental sound*, the *video* stream is analysed (previously ignored), audio event detection and cross-modal gating are new core components, output is overlaid alongside the original video, and a formal 300-clip benchmark replaces ad-hoc news clips. Reusable from the earlier build: FFmpeg audio extraction, Whisper ASR wiring, and the modular pluggable-stage architecture.

3 Research Questions (from the proposal)

- Which types of audio information contribute most to scene understanding when presented visually?
- What level of semantic granularity should be used to represent ambient/environmental sounds?
- Can automatically generated visual augmentations improve accessibility beyond subtitles?
- How should semantic audio augmentation systems be evaluated?
- How does the proposed pipeline compare with representative audio-to-visual approaches?

4 Literature Review

This review covers the eight areas named in the proposal. It is deliberately organised around the *role each area plays in the pipeline* so that model choices in Section 7 follow directly from it.

4.1 Audio Event Detection and Tagging (Stage 4)

This is the engine that turns the soundtrack into a set of labelled non-speech sound events — the raw material for augmentation. The field is anchored by **AudioSet** [1], a 2 M-clip ontology of 527 sound classes that underpins nearly all modern systems. Two families dominate:

- **CNN-based tagging — PANNs** [2]. Large-scale pretrained audio neural networks (notably the CNN14 backbone) established strong general-purpose audio features and a then-state-of-the-art mAP of 0.439 on AudioSet tagging. PANNs remain a robust, lightweight, easy-to-run baseline and are ideal for a first implementation on a 6 GB GPU.
- **Transformer-based tagging — AST and BEATs**. The Audio Spectrogram Transformer (AST) [3] applied a pure-transformer patch model to spectrograms. **BEATs** [4] (audio pre-training with acoustic tokenizers) is the current practical state of the art and is used as the baseline encoder in the DCASE 2023/2024 Sound Event Detection challenges. For frame-level *detection* (with time boundaries, not just clip-level tags), CRNN+BEATs and frame-level transformer variants (e.g. ATST-Frame) lead recent DCASE results.

Relevance / choice. We need both *what* sound and *when* (time boundaries) so augmentations can be time-aligned. Recommendation: start with PANNs (CNN14) for fast, dependency-light clip tagging; move to a BEATs-based tagger for quality; use a CRNN/frame-level model only if precise onset/offset timing proves necessary. AudioSet’s 527-class ontology also gives us a ready-made vocabulary for the “which sounds matter” analysis in Stage 5.

4.2 Audio Captioning (Stage 4/5 alternative representation)

Where tagging outputs labels, *automated audio captioning* (AAC) outputs a free-text sentence describing the soundscape (e.g. “a train approaches as birds chirp in the background”). This is a richer, more compositional representation and a natural bridge to an image-generation prompt. The canonical datasets are **AudioCaps** [5] (AudioSet-derived, ~46 k clips) and **Clotho** [6]. Modern AAC systems pair a strong audio encoder with a language decoder: **EnCLAP** [8] (neural-codec + CLAP + LLM) and **SLAM-AAC** [9] (current SOTA on Clotho/AudioCaps, using paraphrase augmentation and CLAP-based re-ranking) are representative. **CLAP** [7] (contrastive language–audio pretraining) is the workhorse embedding underneath much of this and is directly useful to us as an audio↔text relevance scorer in evaluation. **Relevance / choice.** Captioning is attractive as the “description” that feeds image generation, and it is a natural **baseline** (“audio captioning only, no visual output”). It also provides the *audio captioning* baseline the proposal requires.

4.3 Audio Understanding / Large Audio-Language Models (Stages 3–5)

Large Audio-Language Models (LALMs) fold ASR, audio-event reasoning, and captioning into a single instructable model: an audio encoder + a modality connector + an LLM. Key systems: **Qwen2-Audio** [10] (strong general audio understanding; can be prompted for both transcription and sound description), **SALMONN** [11] (speech + audio-event + music encoders on a frozen LLM), and **Audio Flamingo 3** [12] (fully open, long-context, multi-audio dialogue). Evaluation of these models is itself an active area; **MMAU** [13] is a multi-task audio understanding/reasoning benchmark worth citing for how the community measures these capabilities. **Relevance / choice.** A LALM is an appealing way to *collapse Stages 3–4* into one component and to reason about audio directly in language. Trade-off: a single 7B+ LALM is heavier and less individually inspectable than PANNs + Whisper as separate, swappable modules. Plan: keep the modular ASR+AED path as the dependable default; treat a LALM (Qwen2-Audio) as an alternative Stage-3/4 backend and a possible ablation (“LALM vs. specialised modules”).

4.4 Vision-Language and Multimodal LLMs (Stages 2 and 5)

Stage 2 (what is already visible) and Stage 5 (cross-modal fusion) both lean on vision-language models. **Qwen2.5-VL** [14] is the leading open option: native dynamic-resolution vision, explicit video understanding with temporal grounding (it can localise events to video segments), and 3B/7B/72B sizes — the 7B fits our hardware budget and the 3B is a fallback for the local GPU. **InternVL3** [15] and the **LLaVA** [16] family are the principal alternatives. These same models double as the **VLM describer** in evaluation (Section 10). **Relevance / choice.** Qwen2.5-VL-7B for video understanding and as the evaluation describer; Qwen2.5-VL-3B or a frame-sampled call on the local GPU as fallback. Because the same model family spans understanding *and* evaluation, we should use *different* model instances/prompts for generation vs. judging to avoid a model grading its own output.

4.5 Audio-to-Image Generation (Stage 6 — and a key baseline)

Two philosophies exist. (1) **Direct audio-to-image:** **Sound2Scene** [17] learns to map audio into the latent space of a pretrained image generator and synthesises a scene directly from sound, trained only on unlabelled video; **SonicDiffusion** [19] conditions a pretrained diffusion model on audio. These are exactly the proposal’s “direct audio-to-image” **baseline**. (2) **Text-conditioned generation** driven by a language description of the sound: **Stable Diffusion XL** [20] and **FLUX.1** [21] are the open state of the art. Our proposed method sits in camp (2) but with the crucial extra step that the *prompt is chosen by cross-modal reasoning* (Stage 5), not produced blindly from audio. **Relevance / choice.** Use SDXL / FLUX.1 for augmentation generation (text-conditioned on a Stage-5 prompt); use Sound2Scene as the direct-audio-to-image baseline. This contrast — reasoned, gap-aware prompting vs. blind audio→image — is precisely what the evaluation must isolate.

4.6 Audio-to-Video Generation (Stage 6 stretch goal)

Extending static augmentations to short motion clips is the proposal’s stretch. Directly audio-conditioned video generation is still maturing; the pragmatic route is *image-to-video* applied to a generated keyframe (e.g. Stable Video Diffusion-style motion), keeping the reasoning in Stages 2–5 and treating motion as a rendering choice. This keeps the stretch goal cheap and optional and avoids depending on a fast-moving, heavyweight research area.

TempoTokens (Yariv et al. [18]) — **highly relevant; co-authored by I. Schwartz.** This is a direct audio→video method: a *lightweight adapter* (“AudioMapper”, an MLP) maps **BEATs** audio embeddings into pseudo-text-tokens that condition a *frozen ModelScope* text-to-video diffusion model, so the video is aligned to the audio both globally (semantics) and

temporally (per-segment); it can be conditioned on audio, text, or both. Only the adapter is trained — the T2V model and audio encoder stay frozen — which fits this project’s “glue frozen foundation models” ethos. Trained/evaluated on Landscape, AudioSet-Drums and VGGSound; introduces the **AV-Align** metric (energy-peak alignment between audio and video). Code + pretrained models released (project “TempoTokens”). **How it helps us — three concrete ways:**

1. **As a baseline (primary use).** It is the audio→*video* analogue of Sound2Scene: it generates footage *from audio alone*, with *no* analysis of the existing video and *no* gap-aware gating. That makes it an ideal “direct audio-to-video” baseline to contrast against our cross-modal method (Section 10).
2. **As the Stage-6 motion stretch backend.** If we pursue moving augmentations, it is an off-the-shelf, frozen-model audio-to-video generator we could plug in.
3. **Its AV-Align metric** is directly reusable to score temporal alignment of our augmentations with the sound (Section 10).

It shares components with us (BEATs, VGGSound). **What it does *not* do:** it replaces/creates footage rather than augmenting an existing video alongside it, and it never reasons about what is *already visible* — i.e. it does not perform the cross-modal gating that is our core contribution. So it is a component/baseline, not the whole system. Compute is non-trivial (ModelScope T2V + adapter, GPU; temporal conditioning capped at 24 frames), so it is university-GPU / stretch territory. **[TODO–Adam: decide TempoTokens’ role: (a) the direct audio-to-video baseline (recommended, low effort, strong contrast), and/or (b) the motion-generation stretch backend; check its licence/compute before committing.]**

4.7 Accessibility Technologies for DHH Users (motivation & evaluation grounding)

This literature justifies the whole project and shapes what “useful” means. Two threads matter:

- **Limits of captioning for non-speech sound.** Standard captioning represents only a narrow slice of the soundtrack; substantial atmospheric and emotional information is lost, and non-speech sounds are inconsistently or poorly captioned [22]. Work on visualising prosody/emotion in captions [23] shows the community actively augmenting text with visual cues — directly adjacent to our idea, but for speech rather than ambient sound.
- **Sound-awareness systems.** Jain et al.’s **SoundWatch** [24] (smartwatch sound classification/alerts) and **HomeSound** [25] (in-home sound visualisation, deployed with DHH households) establish, from field studies, *which* sounds DHH users want to be aware of and how visual sound feedback is received. **ProtoSound** [26] personalises sound recognition for DHH users. These ground our choice of “which sounds matter” and provide real user-centred motivation and evaluation framing.

Relevance. This is the section that separates the project from a generic multimodal demo: it names the accessibility gap (non-speech sound), cites evidence that it is real and unmet, and gives a principled basis for the “which audio cues are worth showing” decision in Stage 5.

4.8 Benchmark Datasets

The proposal names three sources for the ~300-clip benchmark:

- **VGGSound** [40] — ~200 k 10-second YouTube clips, 309 classes, each with a *visible* sound source. Excellent for clean single-event ambient/environmental scenarios; its “sound source is on screen” property is useful both as positive material and, inverted, to construct “sound source *off* screen” cases that stress cross-modal gating.
- **UnAV-100** [41] — ~10.8 k untrimmed videos, 30 k+ audio-visual events, 100 categories, avg 2.8 (often overlapping) events per video. Ideal for *mixed audio scenes* and for cases with

multiple concurrent sounds. CC-BY 4.0.

- **MovieNet** [42] — 1,100 movies with rich annotations (42 k scene boundaries, character/place/action/cinematic style tags). Best source for cinematic *acoustic-event* scenes (sirens, glass, explosions, applause) with narrative context.

Selection strategy (decided 2026-08-07 — hybrid, pilot-first, assisted labeling).

The benchmark’s ground truth is the gating judgement itself: per non-speech sound, *is its source visible on-screen?* Clips are categorised **positive_offscreen** (salient sound, source not visible → system should augment), **negative_control** (sound present but source visible → show nothing), or **mixed**; speech-only clips are excluded. Ground truth is *human-labeled* from extracted keyframes, independent of the detector (so PANNs’ misses do not bias it). Sourcing is hybrid: **hand-curated CC clips** (Wikimedia Commons: protests, streets, transport, weather/nature, emergency) for off-screen positives; **UnAV-100** (CC-BY) for mixed/overlapping scale; **VGGSound** (HF mirror) for negative controls, since it is built so the sound source is *visible*. **MovieNet is dropped for clips** — it ships no raw video (registration + agreement, movies excluded). Composition: pilot ~30 → ~120 (fallback) → ~300 (stretch). Tooling + labels live in `benchmark/` (`curate.py`, `manifest.json`, `ATTRIBUTIONS.md`); clips are gitignored.

[TODO–Adam: decide whether background music is a salient augmentation target or excluded (currently flagged, non-salient) — ties to task A6.] [TODO–Adam: confirm the ~300-clip target is feasible within the timeline, or settle on the ~120 fallback.]

5 The Accessibility Gap: Ambient Sound vs. Subtitles vs. Sign Language

This section is the motivational core of the project: *what* non-speech audio information DHH viewers miss, and *how* the two dominant accessibility channels — subtitles and sign language — handle (or fail to handle) it. It is the evidence base that justifies adding a visual channel.

5.1 What ambient/non-speech sound do DHH users miss and want?

Field studies of sound-awareness systems give a concrete, user-grounded answer. Jain et al.’s **HomeSound** [25] and **SoundWatch** [24] were built around a set of ~19 everyday sounds (~31 h of data) that DHH users said they wanted to know about: door knocks/doorbells, a microwave or appliance beeping, a baby crying, a dog barking, a cat meowing, running water, alarms, sirens, a running vehicle, birds chirping. Two observations matter for this project:

- **These are mostly alerting / functional / safety sounds** — things that prompt an action or signal danger in the user’s real environment. That is a different need from *media consumption*, where the missing sounds are often *atmospheric and narrative* (rain setting a mood, an off-screen crowd, ominous music, distant thunder). Existing sound-awareness tech targets the former; this project targets the latter, comparatively under-served niche.
- **The taxonomy still transfers.** The categories — environmental/ambient, discrete acoustic events, human/animal sounds — line up with the proposal’s three benchmark scenarios and with the AudioSet ontology used in Stage 4. **[TODO–Adam: when defining the benchmark’s “which sounds matter” labels, reuse this user-validated taxonomy rather than inventing one.]**

5.2 How subtitles / SDH handle non-speech sound

The relevant caption type is **SDH** (Subtitles for the Deaf and Hard-of-hearing), the most complete form, which — unlike plain subtitles — is expected to represent non-dialogue audio: sound effects, music, speaker identity, manner of speaking, and audience reaction. Conventions are codified in style guides such as the **DCMP Captioning Key** [27]: sound effects are set in square brackets, e.g. `[applause]`, `[glass shatters]`; music is marked with note symbols. *Its*

guidance is strikingly close to this project’s core idea: “the source of the sound effect should be included ... **unless the source is clearly seen on screen**” and “if it can be inferred through the visuals, it is not necessary to caption it.” In other words, **professional captioners already apply cross-modal gating by hand** — describe the sound only when it is not already visible. Stage 5 of this project *automates a principle the captioning industry endorses but performs manually*. This is a strong justification to foreground.

Where SDH falls short (the opportunity):

- **Low bandwidth / lost granularity.** A rich soundscape is compressed to a terse text label. A University of Sheffield study found that in *Jaws*, captions told viewers there was “famous music” but *not* that the music signalled the shark approaching — so DHH viewers lost the suspense the score was built to create [28].
- **Emotional/atmospheric information is lost.** An EEG study found that attending only to standard compliant captions loses substantial emotional information, and that non-verbal-sound captions tend to trigger extra *attentional* effort rather than an emotional response [22].
- **Divided attention.** Reading captions competes with watching the scene; DHH viewers split visual attention between two sources, causing fatigue and missed on-screen content (visual gags, who is speaking) [29]. **[TODO–Adam: design risk to validate: our augmentations are *also* visual, so they must not *worsen* divided attention — placement/timing/salience of augmentations is a real design question and a candidate evaluation dimension.]**
- **No spatial or continuous information.** SDH cannot easily convey *where* a sound is (off-screen left? approaching?) or how it evolves — exactly the information a visual augmentation can carry.
- **Inconsistency.** Non-speech captioning quality varies by provider/regulation; recent HCI work is actively trying to enrich it (visualizing speech style [23, 31]; adapting non-speech captions generatively — CapTune [30]), which shows the community sees the same gap this project addresses.

5.3 How sign language handles non-speech sound

Sign language can carry *more* than subtitles: skilled interpreters convey music and sound through classifiers, mimed sound sources (air-guitar, an instrument), sustained holds for long notes, rhythm, and **non-manual markers** (facial expression for pitch/intensity), aiming to transmit the *feeling* of the sound, not just its name — as seen in theatrical and concert interpreting. Limits for this project’s setting:

- **Availability.** Interpretation is live, human, and costly; the vast majority of recorded/broadcast media has no interpreter, and where present it centres on *speech* (and sometimes music), rarely systematic *ambient/environmental* sound.
- **Automation is unsolved.** Automatic sign-language *generation* remains an open research problem (noted in the proposal’s related work), so it is not a deployable channel for ambient sound today.
- **Takeaway.** Sign language shows the *ceiling* — rich, expressive, affective conveyance of sound — that terse SDH text does not reach. A generated *visual* augmentation is a pragmatic, automatable middle path: higher-bandwidth and more atmospheric than a bracketed label, without requiring a human interpreter or solved sign-language synthesis.

5.4 Synthesis — why this project

DHH viewers miss real semantic and atmospheric information in non-speech sound. Subtitles/SDH capture it only as low-bandwidth text that loses granularity, emotion, spatial and temporal nuance, and adds reading/attention load; and they already *manually* decide to describe

a sound only when it is not visible — the very gating this project automates. Sign language conveys sound richly but is unavailable at scale and not automatable for ambient sound. This positions automatically generated, gap-aware *visual augmentation* as a complementary channel that is (a) higher-bandwidth than SDH text, (b) automatable unlike interpretation, and (c) grounded in a principle captioners already endorse. **[TODO–Adam: frame the report’s introduction around this three-way comparison — it is the clearest articulation of the contribution.]**

5.5 Divided attention: design constraints for augmentation placement

Because our augmentations are *visual*, they compete for the same visual channel as the video and the captions — the divided-attention problem raised in Section 5 could be made worse, not better, if we are careless. The HCI literature gives concrete constraints:

- **DHH viewers already spend most gaze on text and use peripheral vision for the scene.** An eye-tracking study found viewers fixate captions ~84% of the time and watch the video peripherally [32]; deaf viewers’ peripheral vision is more attention-sensitive than hearing viewers’. *Implication:* augmentations can live in the periphery and still be perceived — they need not seize central gaze.
- **Bottom-center placement forces costly gaze switching.** Subtitles at the bottom cause frequent focus switches and raise cognitive load; work like **LightSub** [33] explicitly reduces eye movement. *Implication:* pick a stable, predictable augmentation zone and avoid making the eye ping-pong between scene, caption, and augmentation.
- **Concurrent visual sources fragment attention.** Presenting many visual materials at once fragments perception and raises load [34]. *Implication:* this is a direct argument for aggressive gating — show *few*, well-chosen augmentations, not a busy stream. The core contribution (deciding when to stay silent) is also the main lever for keeping cognitive load acceptable.
- **Referential/spatial cues reduce load.** A gaze/pointing cue toward the referent lowers processing effort. *Implication:* an augmentation that indicates *where* an off-screen sound comes from (e.g. a directional cue) may help more than a free-floating picture.

[TODO–Adam: treat augmentation placement/timing/salience as an explicit design variable and a candidate evaluation dimension — e.g. compare a fixed peripheral panel vs. a spatially-anchored cue, and check it does not increase missed on-screen content. This is a genuine research sub-question, not a cosmetic choice.]

6 Designing Stage 5: Cross-Modal Gating (the core)

Stage 5 decides, per detected sound, *whether* to augment (is it meaningful *and* not already visible?) and *how* (what to depict). This is the project’s novelty, so its design deserves explicit treatment.

6.1 Worked example on real data (validates the premise)

A CC-licensed London NHS-protest clip (Wikimedia) was run through Stage 4. At ~5.5s PANNs detects *Vehicle* (0.55), *Truck*, *honking* and *air horn*; the video frame at that instant shows only the marching crowd and a storefront — **no vehicle is visible**. This is exactly the target case: an ambient sound whose source is off-screen, detected in audio but absent from the picture, and therefore lost to a Deaf viewer relying on the visuals. It concretely validates both the detector and the “augment what is heard but not seen” premise — and shows why Stage 2 (what *is* visible) is needed to make the gating decision rather than augmenting every sound.

6.2 The decision, decomposed

For each candidate audio event e (from Stage 4, with time span) the stage answers:

1. **Salience:** is e semantically meaningful for scene understanding (vs. incidental noise)? Grounded in the user-validated sound taxonomy (Section 5).
2. **Visual redundancy (the gate):** is e 's source/meaning *already visible* in the scene (Stage 2)? If yes $\rightarrow$ stay silent (mirrors the DCMF “don’t caption what’s on screen” rule).
3. **Depiction:** if augmenting, produce a generation prompt + timing + (optionally) a spatial hint for placement (Section 5.5).

6.3 Two signal sources for the gate — and why to combine them

- (a) **LLM reasoning over structured evidence.** Feed a text LLM the Stage-2 visible-object list, the Stage-3 transcript, and the Stage-4 event labels, and ask it to reason about redundancy and salience with structured output. Flexible and explainable, but LLMs *hallucinate* in multimodal reasoning — “more thinking, less seeing” amplifies it, and models over-attend to one modality [38, 39]. **[TODO–Adam: mitigate: ground the decision in explicit structured inputs (labels/lists), force structured JSON output with a required “reason” field, and keep the LLM from inventing sounds/objects not in its inputs.]**
- (b) **Direct on-/off-screen localization.** There is a mature line of work on whether a sound’s source is visible: sound-source localization via cross-modal alignment [35], object-aware localization through audio-visual scene understanding [36], and explicit on-screen-vs-off-screen probability estimation / off-screen separation [37]. This yields a *quantitative* “is the source on-screen?” signal — exactly the gate — without relying on an LLM’s judgement.

Recommendation: use (b) as a grounding signal that *feeds* (a). The localization model proposes an on/off-screen likelihood per event; the LLM then makes the final salience+depiction decision conditioned on it. This reduces hallucinated gating and gives the core mechanism a defensible, measurable basis (and a natural ablation: LLM-only vs. LLM+localization).

6.4 Architecture options

- **Modular (recommended default):** Stage 2 (VLM) + Stage 3 (Whisper) + Stage 4 (PANNs/BEATs) + optional localization $\rightarrow$ Stage 5 (text LLM). Maximally inspectable and ablatability; each signal can be evaluated alone.
- **Omni single-model:** a model that ingests audio+video+text jointly (e.g. Qwen2.5-Omni / Video-SALMONN-class) could collapse Stages 2–5. **[TODO–Adam: consider as an ablation/upper-bound, but verify such a model can be run within our hardware budget and that its decisions stay inspectable; do not make it the default.]**

6.5 Sound granularity requirements (2026-08-20)

Decided from a measurement over 16 real benchmark clips (PANNs at our thresholds): raw salient events fire at 8.2/clip, but after family consolidation and the display threshold this collapses to **0.9** canonical sources/clip (max 2, never 3+ simultaneous); median event span 1.6 s with 37% under 1 s; without consolidation, sub-labels would duplicate massively (Vehicle-family fired 28 times, Bird 17, Rain 11 across the sample).

Taxonomic: the “depictable source” level. Work at the granularity of a nameable, depictable real-world source (Dog, Vehicle, Train, Bird, Siren, Rain) — *not* AudioSet leaves (“Yip”, “Clickety-clack”: undepictable, unreliable) and *not* superclasses (“Animal”: too vague to depict or gate). This is the level at which (a) DCMF says to name a sound source, (b) CLIP can check visibility, (c) retrieval/generation has a subject. Split a family only when depiction *and* the visible concept both differ (Siren $\neq$ Vehicle; Thunder $\neq$ Rain). The FAMILY map in `src/labels.py` implements this and the data shows it is load-bearing.

Temporal: detect fine, display coarse. Keep detection at PANNs’ frame resolution (min span 0.2 s), but never *display* below glanceability: minimum display duration ≈ 2 s (median event is only 1.6 s; 37% would otherwise flash). v1 currently merges all same-source firings into one span (coarsest option) — acceptable for gating evaluation. **[TODO–Adam: Adam: decide gap-aware splitting (e.g. split a source’s span at silent gaps >3 s) if the single-span display feels dishonest on longer clips.]**

Simultaneity: cap at 3 slots. Max observed simultaneous sources is 2; divided-attention evidence (Section 6) says fewer is better. Requirement: panel supports at most 3 slots; overflow drops the lowest-confidence source.

Intensity: continuous, no levels. Confidence $\rightarrow$ opacity is already continuous; the benchmark ground truth needs only binary salience, not loudness levels.

Benchmark ground truth at the same level. Label GT events with the canonical-source vocabulary (the FAMILY parents), binary `source_visible`, spans at ~ 1 s precision — finer timing does not change gating accuracy and would waste labeling effort.

4 of the 16 sampled clips had zero sources above the display threshold. **[TODO–Adam: Adam: after tagging, check those against your labels — if you tagged them unseen/seen ambient, the display threshold (0.15) is slightly too strict and should be tuned on the benchmark.]**

6.6 Annotation guidelines and labelling dilemmas

These questions arose while hand-labelling the benchmark. They are recorded here so the labels stay internally consistent, so the protocol can be stated in the thesis, and because several are genuine design decisions rather than bookkeeping.

The redundancy test (the central rule). The question is *not* “is the sound’s physical source visible in frame?” but “**does the picture already tell the viewer that this sound is happening?**” If it does, the sound is redundant and must not be augmented. Worked example: a rider filmed on a moving motorcycle with wind roar on the soundtrack. No “wind” object is visible, yet the visible speed and motion already convey it, so it is labelled **seen**. Likewise waves when the sea fills the frame, visibly falling rain, and crowd noise over a visible crowd. This is exactly the DCMF captioning rule (“do not caption a sound if it can be inferred from the visuals”, Section 5) that Stage 5 automates, so the ground truth stays aligned with professional captioning practice rather than a naive object-detection notion of “visible”.

Microphone artifacts are not events in the world. Wind noise from airflow over the microphone (riding, cycling, a handheld camera in a breeze) is a recording artifact, not an environmental sound with a depictable source. AudioSet’s **Wind noise (microphone)** is already non-salient; bare **Wind** on a motion clip is usually the same artifact and caused three false augmentations in the first benchmark evaluation.

Broadband noise is systematically mis-labelled, and it dominates the error profile. **Wind**, **Rain**, **Vehicle** and machinery are all wide-band, low-structure textures, and PANNs confuses them with each other and with plain recording noise — at the low operating threshold the detect-everything policy requires, hiss on a mediocre microphone is regularly reported as rain or distant traffic. Consequences: (a) a large share of *suggested* unseen/mixed clips are spurious, so the annotator must judge from the clip and ignore the hint; (b) the same errors are the gate’s errors, making this the single largest accuracy limitation of the v2-(a) pipeline; (c) it motivates the VLM upgrade and a higher evidence bar for the noise-like classes specifically.

Machinery heard as “Vehicle”. Industrial machinery, engines and power tools are routinely reported as **Vehicle**. Because the two are barely separable acoustically at this model scale, the fix belongs on the *visibility* side rather than the detection side: the Stage-2 concept for **Vehicle** also covers visible machinery and equipment, so a machine on screen correctly

suppresses the augmentation instead of producing a phantom off-screen car.

Music is out of scope, including when it is the only “off-screen” sound. Film scores and soundtrack instruments are not ambient sound to depict (decision A6). This bit repeatedly: PANNs reported **Saxophone**, **Music of Asia**, **Strum** and **Double bass** as salient off-screen sources on scored documentary and film footage, producing clips that looked like strong “heard but not seen” cases but were pure soundtrack. Such clips were removed and the labels folded into the music sets. The real boundary is *diegetic vs. non-diegetic*, not “musical”: a busker playing in the scene is a genuine source, though still out of scope for v1.

Faint background sources with no narrative weight. Birds behind a passing train, distant traffic under a conversation. These are genuine non-speech ambient sound and are labelled as such; two mechanisms stop them cluttering the output instead of a special label: the display threshold (Section 6.5) and the three-slot panel cap. The ground truth records what is *there*; the system decides what is *worth showing*.

Trust the clip, not the suggestion. The tagger shows the current pipeline’s suggested label, which is wrong often enough to matter. The label must follow what the annotator hears and sees. These disagreements are useful in themselves: they localise detector errors, and the **Vehicle**/machinery confusion above is the largest single error cluster.

“Mixed” is the selective-gating case, and the scarce one. A clip is mixed when at least one salient source is visible *and* at least one is not (a visible dog barking on a street with unseen traffic). It is the category that demonstrates the system augments *selectively* rather than augmenting everything — the core claim against blind audio-to-image baselines — and it is the hardest to source, hence a dedicated pass.

Three distinct ways to exclude a clip. **BAD** = technically defective (no audio, no video, inaudible, still image, near-duplicate); a written reason is required and retained as a defect record that filters future sourcing. **Drop** = a fine clip that is simply not a case the benchmark needs (another **seen_ambient** when that class is full); set aside in **_dropped/**, not deleted, and excluded from the metric. **I don’t know** = deferral, not a final state.

Clip admissibility, learned from the BAD tags. Every candidate must have both a video and an audible audio stream (mean volume above roughly -50 dB), last at least 15 s, show real motion (no still-image “ambience” uploads), be in colour, come from a distinct source video (no two clips from one video, no adjacent segments), and not be so dominated by added music that the ambient layer is inaudible. These are enforced automatically before a clip reaches the tagging queue.

Never select benchmark clips by the model’s own prediction. A sourcing pass that kept only candidates the gate already classified as the wanted category was abandoned: choosing the test set with the system under test is circular and inflates the reported accuracy by construction. The inclusion criterion is model-independent — “contains a detected non-speech sound” — and the human tag alone determines the category.

6.7 Evaluation hooks specific to Stage 5

The gating decision needs its own metric: on a hand-labelled subset, measure *gating accuracy* — correctly staying silent when the source is visible, correctly augmenting when it is not (precision/recall on the “augment” decision). This directly measures the contribution and is independent of downstream image quality.

6.8 Rendering improvements grounded in the DHH-visualization evidence (2026-08-19)

Follow-up research on non-text sound visualization for DHH users (details in the literature review, §“Beyond subtitles and sign language”) yields concrete, train-nothing upgrades to our *presentation* layer, ranked by evidence strength and cost:

1. **Two-level rendering: glance layer + detail layer.** Pattern/icon recognition is faster than reading, and colour registers in peripheral vision; deaf viewers’ peripheral vision is unusually acute. Show a small persistent *icon* per active off-screen sound (glanceable), and the rich retrieved/generated *image* only for high-salience events. Cheap: we already have the label $\rightarrow$ icon set is a static asset.
 2. **Stable slots, no popping.** Eye-tracking work on dynamic (speaker-following) captions found moving text *distracts* DHH viewers; static placement is preferred. Keep each sound family in a fixed panel slot for the clip’s duration; fade in/out, never jump.
 3. **Loudness $\rightarrow$ visual weight.** SoundVizVR encodes loudness/duration; Fortnite encodes distance as opacity; deaf users preferred displays with amplitude+position over spectral detail. Map per-frame event confidence/RMS to icon size or image opacity — one compositor change.
 4. **Direction from stereo panning.** A pointing cue helps DHH viewers link a sound to a place. For stereo soundtracks, per-channel RMS difference gives a free left/right estimate — render a small arrow on the augmentation. **[TODO–Adam: Adam: many benchmark clips may be mono; check what fraction of the benchmark has usable stereo before investing.]**
 5. **Customization presets.** The strongest DHH finding is heterogeneity (“Choices? That’s the dream”): offer `off` / `icons-only` / `full-imagery` modes — a config flag, and a natural ablation axis for the user study.
 6. **Onset pulse.** Peripheral vision is contrast/motion sensitive: a brief pulse at event onset, then static, draws attention without sustained distraction.
- Items 2, 3 and 5 are one-file changes to the Stage-6 compositor/config; item 1 needs an icon set (e.g. CC icons per sound family); item 4 is a small analysis utility.

7 Model and Tooling Recommendations

Shortlist per stage, with a “local-first” option that fits the GTX 1060 (6 GB) and a “quality” option for the university GPU.

Stage	Local-first dev	(6 GB)	/	Quality (uni GPU)	Licence
1. Audio extract	FFmpeg			FFmpeg	permissive
2. Video under- standing	Qwen2.5-VL-3B sampling)	(frame	Qwen2.5-VL-7B / In-	ternVL3	Apache-2.0*
3. Speech recog- nition	faster-whisper (small/medium)			Whisper Large V3	MIT/Apache
4. Audio events	PANNs (CNN14)			BEATs tagger	MIT / re- search
5. Cross-modal analysis	Llama 3.1-8B / Qwen3-8B (quant.)			Qwen3-14B+ or hosted API	varies
6. Generation	SD 1.5 / SDXL-Turbo (test)			SDXL / FLUX.1-dev	CreativeML/Apache*
7. Evaluation	CLAP + CLIP scores			VLM describe + LLM judge	—

*Verify each model’s exact licence at integration time; FLUX.1 has `dev` (non-commercial) vs. `schne1l` (Apache-2.0) variants, and some audio-event checkpoints are research-only.

[**TODO–Adam: confirm the university GPU VRAM once granted — the split between “local-first” and “quality” columns above depends on it.**] [**TODO–Adam: confirm an acceptable FLUX.1 licence variant (dev non-commercial vs. schnell Apache-2.0) for an academic deliverable; check with the supervisor whether a non-commercial model is acceptable.**]

Hardware reality check. The GTX 1060 (6 GB) can run FFmpeg, faster-whisper (small/medium), PANNs, and SD 1.5 / SDXL-Turbo for development, plus quantised small LLMs slowly. It *cannot* comfortably run SDXL/FLUX at quality, a 7B VLM, or a 14B+ LLM. **Action: request the university GPU early** (lead time) for the quality runs; do all wiring and the benchmark build locally in the meantime. A hosted API for the Stage-5 LLM is a viable fallback if the GPU is delayed.

8 Model Cards — Executive Summaries

A plain-language summary of every model actually wired into the pipeline, so the choices are understandable at a glance. Each card is added when its stage is implemented (research/candidate models live in Section 4).

8.1 PANNs CNN14 (Sound Event Detection) — Stage 4 [*implemented*]

What it is. PANNs (“Pretrained Audio Neural Networks”, Kong et al. [2]) are convolutional networks trained on Google’s AudioSet to recognise 527 kinds of sound. We use the CNN14 “DecisionLevelMax” variant, which produces *frame-level* predictions (a probability for each sound class at each instant), so we know not just *what* sounds occur but *when*.

What it does for us. It is the detector at the heart of Stage 4: given the clip’s audio it outputs, for every ~ 10 ms frame, the probability of each of the 527 AudioSet classes (speech, crowd, siren, dog, rain, applause, ...). We turn those probabilities into (label, start, end, confidence) events and into the timeline heat-map used to “detect everything and plot it”.

How it works (simply). The audio is converted to a log-mel spectrogram (a picture of energy across frequency and time); a CNN reads that picture and scores each sound class over time. It was pretrained on ~ 2 M labelled YouTube clips, so it generalises to everyday sounds without any training on our part.

Why we chose it. Light and fast (runs a 20s clip in < 1 s on CPU — no GPU needed), permissively usable, backed by the standard AudioSet vocabulary, and it gives time boundaries. It is the recommended “first implementation” in Section 7; **BEATs** is the planned quality upgrade later.

Inputs / outputs. In: mono audio (resampled to 32 kHz). Out: a $[\text{frames} \times 527]$ probability matrix $\rightarrow$ event spans + a plot.

Cost / footprint. ~ 327 MB checkpoint (auto-downloaded once to `~/panns_data/`); CPU inference is sub-second for short clips.

Limitations to keep in mind. In speech-dominated clips, background sounds (crowd, cheering) score *low* probabilities, so a low detection threshold (currently 0.05) is needed to surface them — which also lets in more noise. The 527-class ontology is broad but fixed. Frame timing is approximate. These are the reasons the “which sounds matter” decision (Stage 5) is deferred rather than baked into detection.

8.2 CLIP (ViT-B/32) — Stage 2 visibility gate [*implemented, v2-a*]

What it is. CLIP (Contrastive Language–Image Pre-training) is a model that maps images and short text descriptions into the *same* vector space, trained so a picture and its caption land close together. It lets us ask, zero-shot, “does this image match this phrase?” without any task-specific training.

What it does for us. It is the lightweight “what is already visible?” check (Stage 2). We sample a few frames and, for each candidate sound-source phrase (“a photo of a dog”, “... a car on a road”, “... water/a river”), CLIP scores how well the frame matches it (softmax against distractor phrases). Sources that score above a threshold are marked *visible*, and the Stage-5 gate then stays silent on those sounds — so we augment only what is heard but *not* seen.

How it works (simply). Two encoders (one for images, one for text) were trained on ~400 M image-caption pairs to agree; at run time we embed the frame and each phrase and compare by cosine similarity. Sources with no persistent visible object (thunder, explosion, siren) are simply not in the candidate list, so they are never marked visible.

Why we chose it. Small (ViT-B/32, ~600 MB), runs on CPU, no training, and it directly gives a per-source “is it on screen?” signal — exactly the gate. It is the pragmatic stand-in for a full VLM (Qwen2.5-VL), which is the planned upgrade once the university GPU is available.

Inputs / outputs. In: sampled video frames + candidate source phrases. Out: the set of visible sources (with per-source scores), consumed by the Stage-5 gate.

Cost / footprint. ~600 MB checkpoint (auto-downloaded once); a few CPU forward passes per clip. **Limitations.** Zero-shot CLIP is approximate — it can miss a small/partial source or fire on a look-alike (in testing, a bright lab flame read as “Fire” (correct) but also nudged “Bell”); the threshold needs tuning; it judges whole frames, not precise localization. A full VLM or a sound-source-localization model (Section 6) would sharpen this.

8.3 Is AudioSet’s 527-class ontology enough? (coverage assessment)

An honest assessment, because it shapes whether PANNs is sufficient or needs an open-vocabulary companion.

What 527 covers well. It is the de-facto standard audio ontology and covers the common ambient/environmental sounds this project targets. The specific cases raised in discussion are all present: *water flowing* → Water, Stream, Waterfall, Gurgling, Trickle/dribble, Drip, Water tap/faucet; *cat* → Cat, Meow, Purr, Caterwaul; plus Dog/Bark, Rain/Raindrop, multiple Siren types, Wind, Walk/footsteps, glass, applause, machinery. So a sound being “absent from a clip” in our demos reflects the clip’s content, not a blind spot in PANNs.

Three real limitations (not just class count).

1. **Music over-represented.** A large share of the 527 are music genres/instruments, which are mostly irrelevant to ambient accessibility — so the *effective useful* vocabulary is smaller than 527. **[TODO–Adam: decide whether to filter/ignore the music/instrument classes for this task, and quantify exactly how many of the 527 are environmental vs. music (pull the AudioSet ontology and count).]**
2. **Granularity is coarse and uneven.** “Vehicle”, “Water”, “Door” don’t capture direction (a car *approaching from behind*), intensity (gentle stream vs. raging river), or implication — exactly the information most useful to a DHH viewer. Different appliance beeps collapse toward “Beep, beep”.
3. **Closed-set.** PANNs can only emit one of the 527 labels; a genuinely novel or culturally-specific sound is mapped to the nearest class with low confidence, never a new label.

The deeper point. Even a perfect ontology would not suffice, because *a label is a low-bandwidth token* — the same limitation as SDH subtitles ([water], [siren]). The semantic richness in this project must come from Stage 5 reasoning + image generation, not from the detector’s label. The detector only needs to reliably say “water, here, off-screen”.

Open-vocabulary alternative (undecided). If 527 classes prove too restrictive, the fix is an open-vocabulary detector: **CLAP** [7] (match audio against *any* text description, zero-shot) or an **audio-language model** / captioner (Qwen2-Audio [10]) that describes sounds in free text. This is both a robustness upgrade and a legitimate research comparison. **[TODO–Adam: decide whether to add an open-vocabulary path (CLAP or audio-LLM) as a comparison against closed-set PANNs tagging — potentially a headline experiment: “closed-set**

tagging vs. open-vocab captioning for accessibility augmentation”.] [TODO–Adam: the current detection threshold (0.05) and min-duration (0.2 s) are provisional — tune them on real ambient clips once the benchmark exists.]

Concrete limitation found on real data (2026-08-07). A 105 s CC-BY-SA recording of the Israeli “Red Color” (Tzeva Adom) rocket-alert siren + impacts (Sderot) was run through Stage 4. PANNs reliably detected the *off-screen impacts* — Explosion (0.34), Burst/pop, Boom, Artillery fire, Gunshot — i.e. the “something was hit but you don’t see it” sounds. But it **missed the alert siren itself**: the “Siren” class peaked at only 0.025 and “Civil defense siren” at 0.006. The regional electronic warning tone does not match AudioSet’s siren classes (which are dominated by emergency-vehicle wails); the alarm tone + Hebrew announcement were read mostly as Speech. This is a textbook motivation for the open-vocabulary path: CLAP or an audio-LLM could match a text description (“rising-and-falling electronic air-raid warning tone”) that PANNs’ fixed 527 classes cannot. [TODO–Adam: critical-alert sounds (air-raid / red-alert sirens, regional alarms) are NOT reliably covered by PANNs — plan for CLAP/audio-LLM open-vocab detection or a small dedicated detector for such safety-critical alerts if they are in scope.]

9 Results and Findings (2026-08-24)

Three findings from the first full evaluation. They are reported here in the form an examiner would check them, because each is more damaging if discovered than if stated.

9.1 The gate does not yet beat its own ablation

On 274 hand-labelled clips (73 positive, 201 negative):

System	Accuracy	F1
Cross-modal gate (CLIP backend)	49.3%	48.0%
Always augment — <i>no gate at all</i>	29.7%	45.8%
Always stay silent	70.3%	0.0%

Two observations, and the second is the important one. Accuracy is dominated by the negative class, so a constant “never augment” predictor scores 70.3% and beats the gate by 21 points; accuracy is therefore the wrong headline metric here. More seriously, **“always augment” is exactly the blind audio-to-image baseline this project claims to beat**, and the gate improves on it by only +0.023 F1. The central contribution is, at this operating point, not distinguishable from not gating at all. The error is one-sided: 71 of 103 *seen_ambient* clips receive a false augmentation, i.e. the gate declares sources invisible when they are plainly on screen.

9.2 Object presence is not event visibility (the substantive finding)

A blind re-adjudication (Section 9.3) recovered, in the annotator’s own words, what the labelling decision actually turns on:

“bells heard not seen — only *building* seen but bell is heard” “alarm ringing and *is seen* but the ‘ring’ is not indicated” “bird *is seen*, not *singing* but is seen. so this is confusing”

The annotator is judging whether the *sound-producing event* is visible. Every model used in Stage 2 — CLIP, SigLIP and OWLv2 alike — answers whether the *object* is visible. A bell tower is on screen while the bell ringing is not; an alarm housing is visible while its ringing is not; a bird is visible while not singing.

This single distinction accounts for all three symptoms at once: the annotator’s own inconsistency on the positive class, OWLv2 “correctly” locating sources that the annotator had called off-screen, and the 71 false augmentations above. It also predicts that better object grounding — a larger CLIP, a detector, a VLM — cannot fix the gate, because the models are answering a different question from the one the task asks. **Contrastive and detection-based vision–language models expose object presence, not event visibility, and the cross-modal gating task requires the latter.**

9.3 Label stability, and what it licenses

60 already-tagged clips (20 `drop`, 20 `seen_ambient`, 20 `unseen_ambient`) were replayed shuffled with the label hidden, the `drop` option removed and a written reason required.

- Self-agreement 78%, Cohen’s $\kappa = 0.60$ — moderate. The instability is concentrated in the positive class: `seen_ambient` was reproduced 19/20, `unseen_ambient` only 12/20. Part of the gate’s measured error is therefore label noise on exactly the category the contribution depends on, which is consistent with the object/event confusion above rather than with careless annotation.
- Of 20 clips originally dropped, only 1 became a positive; 13 became `seen_ambient` and 6 `no_ambient`. The `drop` bucket is therefore genuine negatives, not withheld positives, so the benchmark cannot be completed from material already collected.

9.4 Sourcing yield, and why the benchmark is frozen

Yield of usable positives did not plateau, it declined as filtering grew more elaborate: 22.2% (naive keyword scraping) $\rightarrow$ 19.7% $\rightarrow$ 15.4%, while the unexplained discard rate rose from 0% to 46%. Four independent strategies (Wikimedia/keyword, AudioSet temporally-strong, curated film scenes, UnAV-100 overlap moments) and a three-stage automatic filter all landed in the same 15–20% band. Yet 36% of clips that contain ambient sound at all are unseen or mixed, so the phenomenon is not rare — the acquisition funnel was discarding it. **[TODO–Adam: Adam: the honest reading is that per-clip labelling of a per-event decision is the bottleneck; report the scarcity itself as a dataset finding rather than as a shortfall against the 300-clip target.]**

10 Evaluation Methodology

10.1 Automatic protocol (primary)

Following the proposal:

1. Analyse the original video + audio.
2. Generate the complementary visual augmentation(s).
3. A VLM *describes* the augmented scene (video + augmentation).
4. Derive a *semantic reference* from the original multimodal input via an LLM — i.e. what a hearing viewer would take from the soundtrack that is not visible.
5. An independent **LLM judge** scores the semantic consistency between the augmentation’s described content and the reference.

Design cautions. (a) Use *different* models/prompts for generation and judging to avoid self-preference bias. (b) Complement the LLM judge with cheaper, less gameable signals: **CLAP** audio $\leftrightarrow$ image and **CLIP** [43] text $\leftrightarrow$ image relevance. (c) Report the judge’s agreement with a small human-labelled subset to validate it.

10.2 Scoring an abstention (2026-09-08)

The first pilot run exposed a flaw serious enough to have invalidated the whole comparison. When a system produced *no augmentation at all*, the describing VLM emitted the string **no augmentation was shown**, and the judge — reading it as a weak but not absurd answer — awarded it **2 out of 4**. Four of the six **proposed** records in the pilot were scored this way.

This is not a small calibration error. It cuts in both directions at once:

- on clips where the soundtrack carries information the picture does not, a system that showed nothing was *rewarded* with half marks for a total failure;
- on clips where the sound adds nothing beyond what is on screen, a system that correctly stayed silent — which is precisely the behaviour the cross-modal gate exists to produce — was *denied* the credit it had earned.

Since the gated system abstains far more often than the two baselines by construction, the error was not noise: it applied asymmetrically to the very system under test, and it blurred the one comparison the thesis turns on.

The fix removes the judge from this decision entirely. An empty candidate is scored deterministically in code (`src/stage7_evaluation/protocol.py`, function `judge`): if the reference is the sentinel “*nothing beyond the picture*”, silence was the correct behaviour and scores **4**; otherwise the reference names information the viewer is missing, none of it was conveyed, and the score is **0**. Silence is never partially correct. The rule covers both empty forms, the image systems’ **no augmentation was shown** and the caption baseline’s **no notable non-speech sound**, so all three systems are scored under one convention rather than the judge’s ad-hoc reading of each phrasing.

Because a mean score can now be reached either by describing sounds well or by abstaining where abstaining happens to pay, the results table reports two further columns per system: the fraction of clips on which it showed nothing, and — among those — the fraction where nothing was the right answer. A system cannot post a respectable mean by strategic silence without that being visible.

Related sampling fix. The same pilot reported results for a single scenario. Clips were being taken in alphabetical order and the `mixed/` folder sorts first, so the run measured only mixed-audio clips — the scenario that most flatters the blind audio-to-image baseline, because there the gate suppresses sounds by design. Clip selection is now stratified round-robin across the four human tags under a fixed seed, so a 100-clip run draws 25 from each.

10.3 Metrics

Semantic-consistency score (LLM judge, rubric-based); augmentation relevance (CLIP/CLAP); *gating accuracy* — did the system correctly *withhold* augmentation when the sound is already visible, and add it when not? This last metric is the direct measure of the core contribution and should be built on a hand-labelled subset. For *moving* augmentations, **AV-Align** [18] (energy-peak alignment between audio and video) measures whether the motion is temporally synced to the sound.

10.4 Baselines (required by the proposal)

(1) Direct audio-to-image (Sound2Scene) — no visual analysis; (2) Audio captioning only — text, no visual; (3) Proposed cross-modal method. The comparison isolates the value of reasoning about the audio-visual gap. For the *video* variant, **TempoTokens** [18] is the concrete “direct audio-to-video, no visual analysis” baseline (Section 4).

10.5 Human evaluation (optional)

If time allows: a small qualitative study comparing “subtitles alone” vs. “subtitles + augmentation” on scene-understanding questions. Given the two-month scope and ethics/recruitment overhead, treat this as optional and keep the automatic protocol as the primary evidence.

[**TODO–Adam: decide whether to attempt the human study at all; if yes, check the department’s ethics/IRB requirements and lead time early — recruitment/ethics can be the schedule bottleneck.**]

11 Project Plan (approx. 8 weeks)

Assumes ~2 months, local GTX 1060 for development, university GPU requested in Week 1. Weeks are indicative and overlap; the benchmark build (data-gathering) runs in parallel with implementation.

Week	Focus	Deliverable / milestone
0 (now)	Refine scope; literature review; plan; request uni GPU ; set up fresh repo structure.	This document; GPU request submitted; repo scaffold.
1	Stage 1–3 skeleton (FFmpeg, faster-whisper, and a first VLM call for video understanding); reuse old wiring.	End-to-end stub runs on one clip.
2	Stage 4 audio event detection (PANNs first, then BEATs); produce labelled+timed events.	Sound events extracted on sample clips.
3	Stage 5 cross-modal analysis: the gap-aware “what to augment” component (LLM prompt design).	First real augmentation prompts; qualitative check.
4	Stage 6 generation (SDXL/FLUX on uni GPU) + compositing augmentations alongside video.	First full augmented clips end-to-end.
5	Benchmark build: curate ~100 (scaling to 300) clips from VGGSound/UnAV-100/MovieNet across the three scenarios; record licences.	Benchmark v1 + selection criteria documented.
6	Evaluation harness: VLM-describe + LLM-judge + CLIP/CLAP; implement gating-accuracy metric; wire the two baselines.	Automatic evaluation runs; baseline numbers.
7	Full experiments across the benchmark; ablations (modular vs. LALM; with/without cross-modal gating); error analysis.	Results tables + qualitative examples.
8	Write-up: technical report, polish prototype, finalise deliverables.	Report draft; reproducible code; curated benchmark.

Critical path & risks. (1) University GPU lead time — request in Week 0; hosted API fallback for Stage 5. (2) Dataset access/download volume (VGGSound is large; download only sampled

clips). (3) Scope creep on the human study and audio-to-video — both explicitly optional. (4) Evaluation validity — validate the LLM judge against a small human-labelled subset early.

12 Action Items — Tasks for Adam

Open questions and external actions that only Adam can resolve. These are the inline **[TODO–Adam:]** markers, collected. Tick them off as they are settled and note the outcome in Section 13.

- A1. Compute — submit the university GPU request now.** Lead time is the schedule’s top risk. Confirm the granted **VRAM** on approval; it decides which models run locally vs. on the cluster (Section 7).
- A2. Dataset access & terms — *resolved 2026-08-07*.** MovieNet dropped (ships no raw video). Hybrid sourcing adopted: curated CC (Wikimedia) for off-screen positives, UnAV-100 (CC-BY) for scale, VGGSound (HF mirror) for source-visible controls (Section 4.8).
- A3. Benchmark size.** Pilot-first agreed (~30 built first). Still confirm whether to push to ~300 or settle on the ~120 fallback given the timeline.
- A4. Model licences.** Confirm an acceptable **FLUX.1** variant (dev non-commercial vs. **schne11** Apache-2.0); check specific **audio-event checkpoints** (some BEATs/DCASE checkpoints are research-only).
- A5. Human study decision.** Decide whether to run the optional qualitative study; if yes, check department ethics/IRB requirements and lead time early.
- A6. Scope of “background music”.** The proposal lists music as a possible audio component but the focus is ambient/environmental sound. Confirm with the supervisor whether music is in or out of scope for augmentation.
- A7. Stages 3–4 design choice.** After a small pilot, decide the default: modular (Whisper + PANNs/BEATs) vs. a single LALM (Qwen2-Audio). Keep the other as an ablation.
- A8. Reference details.** Before the final report, verify the flagged bibliography entries (exact author lists/venues) — see **[TODO–Adam:]** in Section 4/References.
- A9. Security carry-over.** Confirm the previously leaked GitHub PAT was revoked/regenerated (flagged in an earlier session).
- A10. Augmentation placement (design + eval).** Decide how augmentations are shown — stable peripheral panel vs. spatially-anchored directional cue — and treat it as an evaluation dimension that must not increase missed on-screen content (Section 5.5).
- A11. Stage 5 gate signal.** Decide whether to add an on/off-screen *sound-source localization* model as a grounding signal feeding the LLM gate (recommended), and set up the ablation LLM-only vs. LLM+localization (Section 6).
- A12. Open-vocabulary detection (Stage 4).** Decide whether to add a CLAP / audio-LLM open-vocabulary path as a comparison against closed-set PANNs tagging — possibly a headline experiment (Section 8.3).
- A13. Prune the AudioSet vocabulary.** Decide whether to ignore the music/instrument classes for this task; quantify the environmental-vs-music split of the 527 classes.
- A14. Tune detection parameters.** The detection threshold (0.05) and min-duration (0.2 s) are provisional; tune on real ambient clips once the benchmark exists.
- A15. TempoTokens role (audio-to-video).** Decide whether to adopt TempoTokens [18] as the direct audio-to-video *baseline* (recommended) and/or the motion-generation stretch backend; reuse its **AV-Align** metric for temporal alignment. Check licence/compute (Section 4). NB: co-authored by I. Schwartz.

13 Decisions and Notes Log

- **2026-08-22 — Annotation protocol written up** (Section 6.6): the redundancy test (“does the picture already convey it?”, so wind heard while riding is *seen*), mic artifacts vs. real events, broadband-noise mis-labelling as the dominant error mode, machinery-as-Vehicle, music out of scope, BAD vs. Drop vs. unsure, and why clips must never be selected by the model’s own prediction.
- **2026-08-19 — Non-text visualization evidence + rendering roadmap.** Deep-dive on how visuals (not captions/sign) measurably affect DHH users; lit review gained a dedicated section. Derived six evidence-backed presentation upgrades (glance/detail layers, stable slots, loudness → visual weight, stereo-panning direction, presets, onset pulse) — see Section 6. Implemented the three cheap ones the same day: stable per-sound slots (fixed position, dim label when inactive), confidence-weighted opacity, and `RENDER_MODE` presets (`full/minimal/off`); verified end-to-end on the London clip in all three modes, including multi-slot.
- **2026-08-06 — Project pivot confirmed.** Definition changed from “illustrate the speech of news clips” to “augment video with visuals of missing non-speech audio semantics, gated by video understanding.” New definition is authoritative.
- **2026-08-06 — Codebase decision.** Fresh start, reusing pieces (FFmpeg audio, Whisper wiring, modular architecture) from the earlier skeleton.
- **2026-08-06 — First deliverables.** Literature review + project plan (this document).
- **2026-08-06 — Accessibility deep-dive added** (Section 5): what ambient sound DHH users miss, and how SDH subtitles and sign language handle it. Key finding: SDH already performs *manual* cross-modal gating (don’t caption a visible sound source), which Stage 5 automates. Full source trail in the Research Log (Section A).
- **2026-08-07 — v2-(a): CLIP “not-seen” gate working.** Roadmap set by Adam: polish v1 → (a) lightweight gate → (b) diffusion when GPU. Implemented Stage 2 as a lightweight **CLIP** visibility check (model card, Section 8) and enabled the Stage-5 gate. Verified the core premise end-to-end: waterfall (water visible) → shows nothing; dog off-screen → shows a dog; London → shows the off-screen Vehicle but suppresses the on-screen Crowd. Runs on CPU. Next: benchmark gating-accuracy; diffusion generation (b).
- **2026-08-07 — v1 PROTOTYPE working (end-to-end illustrated video).** Decision (Adam): ship a working prototype first, then grow. v1 = detect all non-speech sounds (PANNs) → keep distinct sounds above a display threshold (no seen/not-seen gate yet) → *retrieve* a free CC image per sound (Openverse, no GPU) → composite *alongside* the original video (side-by-side, time-aligned, audio kept). Runs in ~24s on CPU. Verified on the London protest clip: off-screen “Vehicle” sound → real street-traffic photo beside the crowd. The benchmark grind was paused (off critical path). Next (v2): the seen/not-seen gate (Stage 2 VLM) and diffusion generation. Image backend chosen: retrieval now, generation as v2 (Adam).
- **2026-08-07 — Benchmark construction started** (Section 4.8). Decided hybrid sourcing + pilot-first + assisted labeling; dropped MovieNet (no raw video). Built `benchmark/curate.py` (ingest clip → Stage 1+4 → keyframes → draft manifest entry) and a human-labeled `manifest.json`. First 3 clips validate all categories: `london_protest_01` (positive, traffic off-screen), `dog_barking_01` (positive, dog off-screen over a chem-demo), `waterfall_kawaida_01` (negative control, water source fully visible → empty gt).
- **2026-08-07 — Reviewed TempoTokens** (arXiv:2309.16429, co-authored by I. Schwartz). Added to the literature review (audio-to-video), baselines, evaluation (AV-Align metric), references, and Research Log. Best fit: the direct audio-to-video *baseline* + optional motion-stretch backend; it does not do our cross-modal gating. New task A15.
- **2026-08-07 — Red-alert siren MISSED, impacts detected** (Section 8.3). On a real

CC red-alert (Tzeva Adom) recording, PANNs detected explosions/impacts well but scored the alert siren at only 0.025 — a concrete closed-set/training-distribution limitation that strengthens the case for an open-vocabulary (CLAP / audio-LLM) path for safety-critical alerts.

- **2026-08-07 — Off-screen-sound premise validated on real data** (Section 6.1). On a London NHS-protest clip, PANNs detected Vehicle/Truck/horn at ~ 5.5 s while the frame showed only the crowd (no vehicle visible) — a real instance of “heard but not seen”. Also verified detection on a CC dog-barking video (dog cluster fired).
- **2026-08-06 — Stage 4 coverage assessment** (Section 8.3). Confirmed the sounds of interest (water/cat/dog/rain/siren/footsteps) ARE in AudioSet’s 527 classes; documented the three real limits (music over-representation, coarse granularity, closed-set) and the open-vocabulary alternative (CLAP / audio-LLM) as an undecided comparison. New tasks A12–A14.
- **2026-08-06 — Strategy: detect-everything-first.** Decision (Adam): at first, detect *all* ambient sounds (even faint background dog barks, sirens) and visualise them; decide which matter *later*. Implemented: Stage 4 now runs **PANNs CNN14 SED** for real (executive summary in Section 8), with a low threshold (0.05) and a timeline heat-map plot per clip; Stage 5 gate is set to *pass-through* (`GATE_ENABLED=False`) so nothing is filtered yet. Verified on the synthetic clip and a real 20.9 s news clip (surfaced Speech + Crowd, Outside-urban, Basketball bounce, Vehicle, etc.). Next: use video frames to disambiguate confusing sounds (Stage 2 VLM).
- **2026-08-06 — Divided-attention design constraints** (Section 5.5) and **Stage 5 cross-modal gating design** (Section 6) researched and written. Key outcomes: augmentation placement is a real design/eval variable and aggressive gating helps cognitive load; combine an on/off-screen *localization* signal with a *grounded* LLM decision to curb hallucination; keep a modular, ablatabe architecture; measure gating accuracy.
- **2026-08-06 — Constraints unchanged.** ~ 2 months; local GTX 1060 6 GB; university GPU to be requested.
- **Open items.** Confirm university GPU VRAM once granted; finalise exact benchmark composition after a first pass over the datasets; decide modular-vs-LALM default for Stages 3–4 after a small pilot.

[**TODO–Adam: verify the exact author lists/venues/years of the entries marked “[verify]” below before the final report — these were assembled from search results and not all are fully confirmed.**]

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A Research Log — Sources Consulted and Conclusions

A record of the background research behind this document: the topics investigated, the key primary links, and the conclusion drawn from each. Web searches were run in August 2026; where a claim rests on secondary coverage rather than a confirmed primary paper it is tagged for verification.

Audio event detection (Stage 4)

Key links: PANNs <https://arxiv.org/abs/1912.10211>; AST <https://arxiv.org/abs/2104.01778>; BEATs <https://arxiv.org/abs/2212.09058>; AudioSet tagging code https://github.com/qiuqiangkong/audioset_tagging_cnn. **Conclusion:** PANNs/CNN14 is the light, dependable first implementation; BEATs is the quality upgrade and the DCASE baseline encoder; use a frame-level model only if precise onset/offset timing is required. AudioSet’s 527-class ontology is our sound vocabulary.

Audio captioning (baseline + prompt source)

Key links: SLAM-AAC <https://arxiv.org/abs/2410.09503>; AudioSetCaps <https://arxiv.org/abs/2411.18953>. **Conclusion:** captioning gives a compositional soundscape description usable as a generation prompt and doubles as the required “audio captioning only” baseline; CLAP is reusable as an audio↔text relevance scorer in evaluation.

Large audio-language models (Stages 3–4 alternative)

Key links: Qwen-Audio/Qwen2-Audio <https://arxiv.org/abs/2311.07919>; Audio Flamingo 3 <https://arxiv.org/abs/2507.08128>; MMAU benchmark <https://arxiv.org/abs/2410.19168>. **Conclusion:** a LALM can collapse ASR+AED into one instructable component; keep modular Whisper+PANNs as the inspectable default and treat the LALM as an ablation.

Vision-language models (Stages 2 and 7)

Key links: Qwen2.5-VL <https://arxiv.org/abs/2502.13923>; Qwen2.5-VL-7B weights <https://huggingface.co/Qwen/Qwen2.5-VL-7B-Instruct>. **Conclusion:** Qwen2.5-VL-7B for video understanding and as the evaluation describer (3B as local fallback); it has explicit temporal grounding. Use different instances/prompts for generation vs. judging.

Audio-to-image / diffusion (Stage 6 + direct baseline)

Key links: Sound2Scene <https://sound2scene.github.io/> (code <https://github.com/kaist-ami/Sound2Scene>); SonicDiffusion <https://cyberiaada.github.io/SonicDiffusion/>; SDXL <https://arxiv.org/abs/2307.01952>; FLUX.1 <https://blackforestlabs.ai/>. **Conclusion:** generate with text-conditioned SDXL/FLUX driven by the Stage-5 (gap-aware) prompt; use Sound2Scene as the “direct audio-to-image, no visual reasoning” baseline — the contrast is exactly what the evaluation isolates.

Benchmark datasets

Key links: VGGSound <https://arxiv.org/abs/2004.14368> / <https://github.com/hche11/VGGSound>; UnAV-100 <https://unav100.github.io/> / <https://arxiv.org/abs/2303.12930>; MovieNet <https://arxiv.org/abs/2007.10937> / <https://movienet.github.io/>. **Conclusion:** VGGSound for clean ambient/event clips (visible source; invert for “off-screen” cases), UnAV-100 for mixed/overlapping events (CC-BY 4.0), MovieNet for narrative acoustic events. Access/terms need checking (Task A2).

Accessibility — what DHH users miss and how subtitles/sign language cope

Key links: SoundWatch <https://dl.acm.org/doi/10.1145/3373625.3416991>; HomeSound https://soundability.eecs.umich.edu/img/portfolio/Jain_HomeSound_CHI2020.pdf; DCMP Captioning Key <https://dcmp.org/learn/captioningkey>; EEG caption-limitation study <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7040021/>; live-caption UX <https://arxiv.org/abs/2404.10153>; CapTune <https://arxiv.org/abs/2508.19971>; Sheffield suspense/captions coverage <https://liamodell.com/2024/05/09/captions-cinema-suspense-university-of-sheffield-paper->

Conclusions: (1) DHH users’ desired sounds are mostly *alerting/functional* (SoundWatch/HomeSound); media *atmospheric* sound is the under-served niche this project targets. (2) SDH is the only caption type that covers non-speech sound, but it is low-bandwidth text that loses granularity (the *Jaws* suspense example), emotion (EEG study), and spatial/temporal nuance, and adds divided-attention load. (3) The DCMP rule “don’t caption a sound whose source is visible on screen” means professional captioners already perform *manual cross-modal gating* — Stage 5 automates an endorsed principle. (4) Sign language conveys sound richly but is unavailable at scale and not automatable for ambient sound, so it marks the expressive ceiling rather than a deployable channel. Together these motivate gap-aware visual augmentation as a complementary channel.

Audio-to-video generation — TempoTokens (arXiv:2309.16429)

Read on request. Link: <https://arxiv.org/abs/2309.16429>; code: <https://pages.cs.huji.ac.il/adiyoss-lab/TempoTokens>. **Conclusion:** directly relevant (and co-authored by I. Schwartz). A frozen-model audio-to-video generator (BEATs → light adapter → ModelScope T2V) that aligns video to audio globally + temporally. Best used as (1) the “direct audio-to-video, no visual reasoning” baseline — the video analogue of Sound2Scene, contrasting with our gap-aware method; (2) an off-the-shelf Stage-6 motion-stretch backend; (3) its AV-Align metric for scoring temporal alignment. It does NOT do cross-modal gating (never looks at what is already visible), so it complements rather than replaces our contribution. Compute-heavy (GPU; 24-frame cap).

Divided attention & augmentation placement (design constraints)

Key links: captions vs transcripts eye-tracking <https://www.cs.cmu.edu/~jbigham/pubs/pdfs/2013/captionvstranscripts.pdf>; LightSub <https://doi.org/10.3390/mti8060051>; captions+sign coexistence <https://pmc.ncbi.nlm.nih.gov/articles/PMC12922059/>; eye-gaze in sports captions <https://dl.acm.org/doi/10.1145/3806043>. **Conclusion:** DHH viewers fixate text ~84% of the time and use peripheral vision for the scene, so augmentations can sit in a stable peripheral zone; bottom-center placement and many concurrent visuals raise load, which argues *for* aggressive gating (few, well-chosen augmentations) and for spatially-anchored cues. Placement/timing/salience is a real design + eval variable (see Section 5.5).

Stage 5 — cross-modal gating (on/off-screen + grounded LLM decision)

Key links: sound-source localization via cross-modal alignment <https://arxiv.org/abs/2309.10724>; object-aware localization <https://arxiv.org/abs/2506.18557>; off-screen separation <https://pmc.ncbi.nlm.nih.gov/articles/PMC10181533/>; Fork-Merge decoding <https://arxiv.org/abs/2505.20873>; cross-modal attention imbalance <https://arxiv.org/abs/2509.18816>. **Conclusion:** there is a mature “is the sound source on-screen?” research line that yields a quantitative gate signal; combine it with a text LLM reasoning over structured Stage-2/3/4 evidence (not raw media) to curb the known MLLM hallucination/attention-imbalance failure modes. Default to a modular architecture (inspectable, ablable); treat an omni single-model as an ablation subject to the hardware budget. Gating accuracy on a hand-labelled subset is the metric that isolates the contribution (Section 6).